

Kozuchi Agent: A Language-Agnostic Open-Weight Agent for Software Repair

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Abstract

Industrial software-engineering teams increasingly prefer LLM agents that turn bug reports into correct, test-passing patches, but operating such agents at industrial scale raises harness-level problems: long horizons, tool-use discipline, context persistence, multi-cluster execution, and evaluation reuse. This paper presents Kozuchi Agent, a *language-agnostic open-weight agent* and *CI-operated evaluation pipeline* for industrial-scale evaluation of issue-to-patch agents. Kozuchi agent combines explicit phases, persistent intermediate state, deterministic software-engineering tools, model-independent action interfaces, and cross-agent test-time selection, making agent behavior auditable and repeatable without proprietary models or large fine-tuned backbones. With a locally hosted Qwen3.5-27B model, no fine-tuning, and TTS@8 candidate generation, Kozuchi agent resolves 374 of 500 SWE-bench Verified instances on the official cloud evaluator. Re-running the same 27 B agent unchanged on Multi-SWE-bench Java Verified resolves 41/128 instances (32.03%), placing first among strict open-weight submissions and fourth of 42 overall. Per-phase scaffold behaviour matches the Python run within ± 5 percentage points on every phase, evidence that the harness is language agnostic. It ranks 12th among 135 public Python submissions and first among open-weight systems. The analysis shows that remaining failures are dominated by semantic correctness, Java-specific harness failures, and selection errors rather than proprietary-model advantages, while the CI workflow compresses operator-facing touch-points from five to one.

1 Introduction

Across our engineering organization, we observe a recurring industrial demand: developers want an LLM-based agent that takes an issue description plus a repository state and emits a correct, test-passing patch. In SWE-bench-style deployments [18], meeting that demand requires more than choosing a model [48]: the surrounding harness must keep long, tool-using repair attempts auditable and repeatable. The central question for us became: *how far can an open, locally hosted model go when embedded in an auditable repair harness and selected through cross-agent testing?* Four operational blockers shaped the design.

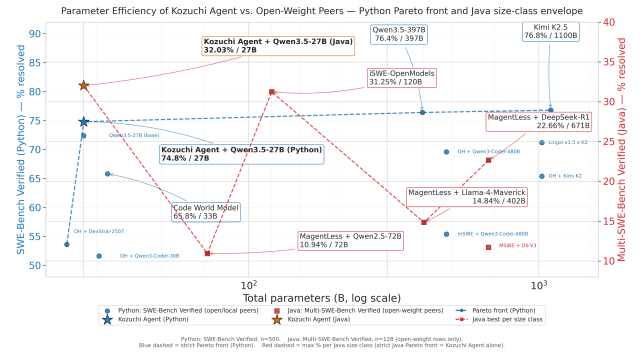


Figure 1: Kozuchi Agent is parameter efficient on both Python SWE-bench Verified and Multi-SWE-bench Java Verified.

Pain 1: long-horizon execution. A real fix requires reproducing the bug, generating a regression test, locating the root cause, editing several files, and verifying the result [18, 34, 42, 44], with each step taking many tens of LLM turns. In practice software-repair benchmark trajectories consume most of a multi-hour budget. Free-form planning at that horizon degrades decision quality: agents skip reproduction, re-investigate locations they analyzed, or “fix” code they have not understood.

Pain 2: tool-grammar drift across model families. LLM families vary in conventions for tool calls and structured actions. A small syntax mismatch can invalidate a reasonable action [4, 31] and cascade into wasted turns. The harness must absorb that diversity rather than forcing every model integration to fork the agent.

Pain 3: multi-cluster heterogeneity. Our jobs land on heterogeneous internal clusters: GPU inference and training, VM-based testing, and Docker-based benchmark grading all have different operational requirements. We need one runtime to work across these environments without forcing researchers to rewrite scripts for each hardware backend.

Pain 4: evaluation cost. Re-running SWE-bench Verified end-to-end with Docker-based grading is expensive enough that a CI pipeline that re-runs every stage is unusable in practice [29, 36]. The

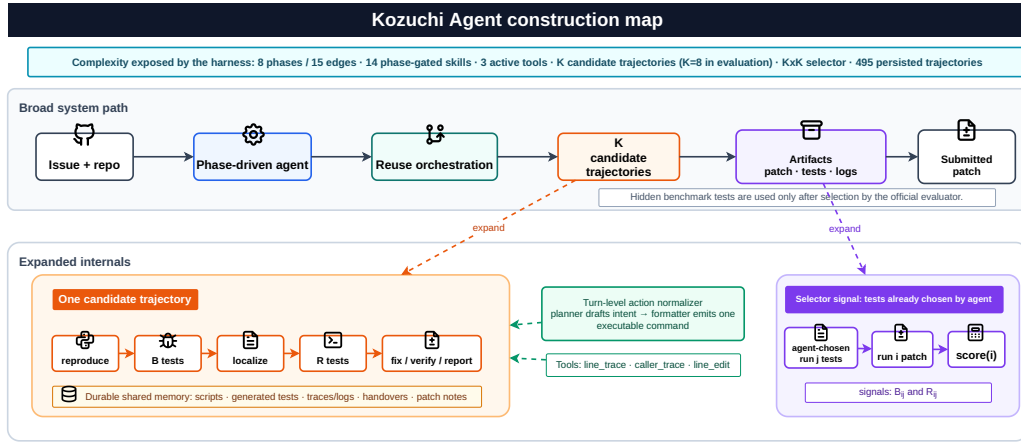


Figure 2: Kozuchi Agent workflow from candidate generation to cross-agent selection.

evaluator-grade score is the gate for external claims and the floor for experimentation, but it must be reachable on a budget that fits a few SLURM jobs [45] rather than a research-cluster reservation. The CI reuse mode declared in `.gitlab-ci.yml` short-circuits 6 of 9 stages per iteration (§9).

This paper is an experience report on the harness and pipeline we built to address those four pains, and on what we learned operating it. *Kozuchi Agent* comprises:

- a phase-driven SWE agent layered on mini-swe-agent, a minimal open-source agent runtime [9];
- a fixed sandbox of dynamic-analysis and edit operations;
- a CI-driven multi-cluster pipeline for repeatable pipelines;
- a test-time selection layer that merges multiple inference runs through *cross-agent testing* [19].

The released artifacts expose the resulting trajectories, selector choices, evaluation outputs, and additional deep analysis.

Contributions. C1. A phase-driven, model-agnostic SWE agent harness with auditable tools. We decompose issue resolution into eight semantically meaningful phases connected by explicit success and fallback transitions, separate the logical action contract from model-specific surface syntax, and expose a small deterministic tool sandbox for dynamic tracing, caller discovery, and guarded editing. Together these choices turn a free-form repair loop into an inspectable workflow whose state can be resumed and compared across open-weight and API-served backends.

C2. A CI-operated multi-cluster benchmark pipeline with reusable and regenerable artifacts. A single CI pipeline coordinates benchmark inference, grading, data synthesis, optional post-training, test-time selection, and reporting across heterogeneous clusters. Its reuse mode allows operators to short-circuit expensive stages using previously validated outputs, while the published analysis scripts in `src/` regenerate every empirical value cited in RQ1–RQ6 directly from the SWE-bench and Multi-SWE-bench Java harness reports, trajectories, and cross-agent selection bundles in the public artifact (§11).

C3. A cross-agent test-time selector with trajectory-level audit. We merge $K = 8$ candidate inference runs through cross-agent

testing: each run contributes both a patch and independent tests, and the tests are cross-applied to score candidate patches without using the hidden benchmark tests or a learned verifier. We also analyze the eight candidate streams and 495 full message trajectories, separating selector regret from instance hardness (408 instances solved by at least one run, 234 by all runs, 92 by none) and surfacing content-level selector signals such as THOUGHT-block-to-action balance, rollback language, and error-marker asymmetries.

C4. Competitive open-weight repair results with statistical and failure-mode analysis. On the official SWE-bench Verified cloud evaluator, our 27B open-weight Python configuration resolves 374 of 500 instances (74.80%, Wilson 95% CI [70.82%, 78.41%]). The internal Docker re-grade resolves 376 (75.2%); the cross-agent selector contributes +14 resolved instances over a size-ordered first-candidate baseline. On a frozen snapshot of 135 publicly catalogued SWE-bench Verified submissions the configuration ranks 12th overall and is the highest-ranked open-weight system; under Benjamini–Hochberg FDR correction (target $q \leq 0.05$) the configuration significantly outperforms 16 of 17 curated open-weight peers on matched per-instance McNemar tests. Of 126 unresolved instances, 91.3% are patches that apply cleanly but do not flip the hidden tests; 0% are malformed-diff or empty-patch failures, isolating the residual gap as a semantic-correctness problem rather than an editing-format problem. Patch LOC churn is the strongest single feature signal (point-biserial $r = -0.197$, $p = 1.1 \times 10^{-5}$), and 494/495 produced patches apply cleanly through the SWE-bench harness. On Multi-SWE-bench Java Verified, the unchanged 27 B Java configuration resolves 41/128 instances (32.03%), ranks 4 of 42 overall, and is first among strict open-weight submissions; under Benjamini–Hochberg FDR control ($q \leq 0.05$), it significantly outperforms 34 of 41 Java peer systems on paired per-instance outcomes.

C5. Cross-language transfer of the same harness to Java. The phase decomposition, action contract, persistent shared state, and cross-agent selector transfer from Python to Java unchanged. Per-phase share of assistant messages matches the Python run within ± 5 percentage points on every phase, six of seven per-instance effort medians sit within 20% of Python parity, and the same 27 B

Qwen3.5 backbone beats the strongest same-class open-weight Java peer (Qwen2.5-72B) by 38 instances, compared with a 26-instance same-family margin on Python.

Roadmap. §2 situates the work among agentic SWE systems, industry-facing coding agents, training pipelines, and LLM-enabled CI/CD systems. §3 sets the SWE-bench, cluster, and prior-workflow background. §4 defines the runtime, trajectory, phase graph, tool-call contract, verifier/selector, and artifact boundary, and lists five invariants the runtime maintains. §5 narrates five empirical observations that shaped the design. §6 describes the system in five layers (runtime, tools, pipeline, serving, CI). §7 details the cross-agent test-time selector. §8 answers six research questions on SWE-bench Verified and Multi-SWE-bench Java Verified, all of them grounded in scripted analyses over the published Python TTS@8 and Java xcheck@8 artifacts: headline accuracy with uncertainty, peer-ranked position with multiplicity-corrected paired tests, structural breakdowns by repository, patch size, and trajectory effort, the failure taxonomy and operational-reliability profile, the TTS@8 selection lift and cost-diversity envelope, and cross-language transfer to Java using the same agent, backbone, and scaffold. §11 gives the artifact availability statement and release boundary for the public trajectories, reports, selector bundle, and analysis scripts. §9 states seven operational lessons. §10 discusses future directions.

2 Related Work

Agentic SWE systems. SWE-agent [42] established the now-standard pattern of an LLM observing a shell and emitting commands, building on broader reason-act and reflective-agent patterns [25, 34, 43, 44]. We build on a minimal open-source runtime and add a phase-driven contract, a fixed tool sandbox, and a multi-cluster pipeline. AGENTLESS [40] uses a simpler localization-fix-verify pipeline; OpenHands and AutoCodeRover provide complementary open software-agent platforms and program-improvement systems [37, 47]. Our system is closer to a long agentic loop but constrains it through phases and shared state. The clearest difference is how test assets are handled: AGENTLESS selects one reproduction test and one regression set per issue, whereas each Kozuchi agent run independently generates tests that are later cross-applied to other candidates.

Industry-facing coding agents. Commercial systems such as GitHub Copilot’s cloud coding agent [13], Amazon Q Developer [2], Google Jules [20], OpenAI Codex [30], Anthropic Claude Code [3], and Devin [39] validate industrial demand for asynchronous issue-to-patch agents. Their public descriptions are necessarily product-oriented; our focus is complementary harness evidence: phase boundaries, deterministic tools, queue-aware CI orchestration, reusable benchmark artifacts, and inspectable selection.

Tools, training, and CI. Toolformer-style work injects tool calls into model training [33]; generated-test selection and pass@k-style code evaluation provide related precedents for using execution evidence around generated code [6, 7]. Our approach is the dual, holding the model fixed while injecting classic SE tools (line tracing, caller discovery, guarded editing) at runtime. SFT and RL recipes for code agents remain active directions, and systems such as Nemotron-CORTEXA [35] use fine-tuned retrievers and graph

localization. R2E-Gym [17] explores verifiers for BEST@K; our cross-agent testing is execution-based but deliberately model-free. Finally, deployed LLM-for-CI systems such as LogSage [41] share our concern for artifact availability and reproducibility, although in our case CI orchestrates the agent workload itself.

3 Background and Industrial Context

3.1 SWE-bench and Multi-SWE-bench Java

SWE-bench is a benchmark of real GitHub issues paired with the affected repository state and a hidden test suite; an entry counts as *resolved* only when the model’s patch passes the issue-revealing tests while preserving the existing regression tests [18]. It extends a longer code-generation and program-repair evaluation line from function-level benchmarks to repository-level issues [7, 12, 14, 22–24]. SWE-bench Verified is a 500-instance human-curated subset with cleaner problem statements and reliable evaluation tests [29]; it is the Python benchmark used in RQ1–RQ5. Multi-SWE-bench Java Verified is the corresponding Java track used in RQ6: it keeps the same issue-to-patch resolved metric, but evaluates Java repositories through Maven/Gradle-style harnesses and a separate public leaderboard [46]. We report the Python result on the sb-`cli` cloud stack [36] and the Java result on the Multi-SWE-bench Java Verified leaderboard, keeping each benchmark’s denominator and peer set separate.

3.2 Agentic loops and base runtime

Modern SWE agents follow an agentic loop: a tool-using LLM observes a shell, plans, runs a command, observes the result, and repeats [34, 44]. We build on a minimal open-source runtime [9] with harness-level support for long repairs: phase control, persistent shared state, deterministic tools, and model-independent action formatting. The artifact appendix defines the release boundary: trajectories, selector choices, evaluation outputs, and deep analysis are public, while agent and pipeline source code are not.

3.3 Multi-cluster operating context

Our infrastructure is heterogeneous and shared across many groups. Candidate generation, grading, and optional training run on different GPU and VM-backed clusters, each with different module systems, container support, storage paths, and queue behavior. The central methodological point is that the pipeline hides those differences behind a single cluster abstraction: a stage describes what work it needs, while the environment layer supplies the local launch details. Concrete cluster names and runner wiring are listed with the released artifact rather than in the main narrative.

3.4 The workflow we set out to replace

Before Kozuchi Agent, each iteration required manually starting a vLLM server, launching a one-off repair-benchmark run, waiting hours, running Docker grading, and recording scores in a spreadsheet. SFT and RL experiments used separate branches and scripts; changing models or comparing trajectories required file edits and manual log reading, which are error prone. The harness replaces this with a CI-driven pipeline, one configurable agent, a fixed runtime contract, and a publishable artifact bundle. Table 1 summarizes the

Table 1: Per-evaluation-cycle workflow comparison.

Aspect	Pre-Kozuchi agent	Kozuchi agent
Operator touch-points / cycle	5 (vLLM, launch, wait, grade, log)	1 (CI push)
Operator effort / cycle (est. min)	≈75	≈5
Backend integrations	separate integration effort	4 action formats × 17 model configs
Reusable pipeline stages	0 of 9	6 of 9
Cluster targets / abstraction	per-cluster scripts	5 (ENV_NAME)
Score provenance	manual score log	495 traj. + selection JSON
Patch applicability	not measured	494/495 clean apply

workflow replacement; the engineer-minute estimate is bounded by published SWE-bench Docker-runtime and CI-adoption studies [1, 15].

4 Definitions and Design Invariants

We adopt the following definitions for the rest of the paper. Some invariants are mechanically enforced by parsing and configured transitions; others are prompt/workflow contracts audited through trajectories.

D1 (Agent runtime $\mathcal{R}$). The runtime is $\mathcal{R} = \langle \mathcal{P}, \mathcal{S}, \mathcal{T}, \mathcal{A}, \mathcal{F}, \mathcal{H} \rangle$: phases, shared state, tools, action format, parser, and handover function. This tuple is declared by configuration rather than recovered from ad hoc scripts.

D2 (Trajectory). A trajectory is the ordered record of prompts, assistant completions, parsed tool calls, tool responses, format-/timeout feedback, phase handovers, and shared-state snapshots. Trajectories are published per instance and are the main audit object.

D3 (Phase transition system). The phase transition system is a directed graph $G_\Phi = (V_\Phi, E_\Phi)$. The evaluated graph has eight phases and fifteen normal/fallback edges (Algorithm 1).

D4 (Tool-call contract). Each assistant turn must contain exactly one parseable action. Four surface formats let the same logical contract serve different model families while keeping parser failures explicit in the trajectory.

D5 (Verifier/Selector). A verifier returns a binary or numeric pass-rate signal for a candidate patch. We use an intra-trajectory verifier for tests generated inside one run, and a cross-agent selector (§7) that cross-applies each run’s generated tests to every candidate patch.

D6 (Artifact boundary). The public artifact includes trajectories, patches, per-run reports, selection tables, and official submission files. Transient scratch state, internal cluster configuration, and pipeline source are outside the release boundary because they contain internal paths, runners, tokens, or licence-sensitive dependencies.

Runtime invariants. Every assistant turn yields one parseable action or a recorded format error; phase exits follow configured edges; generated notes and logs stay outside the submitted patch; long observations are summarized with explicit elision; and repeated failures or repeated identical actions trigger a strategy-change warning.

Algorithm 1 Phase-driven repair orchestration

Input: Issue I , repository checkout R , phase graph Φ , skill library $\mathcal{K}$, tool set $\mathcal{T}$, action grammar $\mathcal{A}$, shared artifact store S , and limits B .
Output: Candidate patch Δ , final report ρ , and trajectory bundle τ .

```

1:  $p \leftarrow p_{\text{reproduce}}$ ;  $H \leftarrow \emptyset$ ;  $M \leftarrow \emptyset$ ;  $\tau \leftarrow \emptyset$ 
2:  $\triangleright$  In the evaluated graph,  $p$  ranges over issue reproduction, FAIL-to-PASS test synthesis, code localization, PASS-to-PASS test localization, code fixing, patch verification, issue closeout, and final reporting.
3: while  $p \neq p_{\text{final}}$  do
4:    $K_p \leftarrow \{k \in \mathcal{K} : p \in \text{phases}(k) \wedge \text{tools}(k) \subseteq \mathcal{T}\}$ 
5:    $C_p \leftarrow \text{Render}(I, R, \Phi[p], K_p, S, M, H)$ 
6:    $\triangleright C_p$  contains the phase definition, available tools, selected skill guidance, handover memos, and artifacts already written to  $S$ .
7:    $e \leftarrow \perp$ 
8:   while  $e = \perp$  do
9:      $y \leftarrow \text{LLM}(C_p, H)$ 
10:     $a \leftarrow \text{Parse}_A(y)$ 
11:    if  $a = \perp$  then
12:       $o \leftarrow \text{FormatError}(y, A)$ 
13:       $\triangleright$  Ambiguous or multi-action output is recorded but not executed.
14:    end if
15:    if  $a \neq \perp$  then
16:       $o \leftarrow \text{SandboxExec}(a, R, \mathcal{T})$ 
17:       $S \leftarrow \text{UpdateArtifacts}(S, p, a, o)$ 
18:       $\triangleright$  Phase artifacts include reproduction scripts, tests, traces, root-cause notes, patch diffs, and reports.
19:    end if
20:     $H \leftarrow H \circ (y, a, o)$ ;  $\tau \leftarrow \tau \circ (p, y, a, o, S)$ 
21:     $e \leftarrow \text{ExitMarker}(y, o)$ 
22:    if  $e = \perp$  and  $\text{OverBudget}(H, S, B, p)$  then
23:       $m \leftarrow \text{Compress}(H, S, p)$ ;  $S \leftarrow S \cup \{m\}$ ;  $H \leftarrow \emptyset$ 
24:       $C_p \leftarrow \text{Render}(I, R, \Phi[p], K_p, S, M, H)$ 
25:    end if
26:  end while
27:   $(p, M) \leftarrow \text{PhaseHandover}(\Phi, p, e, H, S, M)$ 
28:   $\triangleright$  Handover extracts phase artifacts, follows the configured next/fallback edge, writes a memo, and preserves scratch state outside the patch.
29: end while
30:  $(\Delta, \rho) \leftarrow \text{CollectFinalArtifacts}(S, M)$ 
31: return  $(\Delta, \rho, \tau)$ 

```

5 Motivating Empirical Observations

The harness design emerged from operational failures in early prototypes; each observation below maps to a mechanism in §6.

O1 (Long conversations forget). Single-prompt repair attempts skipped reproduction, re-investigated old locations, or patched code they had not understood. We responded with an explicit phase sequence and required the agent to state its workflow position at every turn.

O2 (Tool syntax is a cliff). Different model families emit different tool-call bracketings. Separating the logical action from its surface syntax made model switching a configuration change rather than an agent fork.

O3 (State must survive compression). Evidence stored only in conversation text can disappear when context is compressed. Reproduction notes, generated tests, traces, task boards, and handover memos therefore live in shared filesystem state.

O4 (Shell alone localizes poorly). Root-cause localization benefits from classic dynamic analysis. We added a small deterministic tool

sandbox for line tracing, caller discovery, and guarded editing rather than a large, brittle tool zoo.

O5 (CI cost is repeated work). Benchmark stages take hours. Reuse mode lets operators hold validated artifacts fixed while changing one stage, which made repeated TTS@8 and selector sweeps practical.

6 System Design

Figure 2 summarizes the architecture at two connected resolutions. The broad path shows how the pipeline turns one issue into K auditable candidate trajectories and a final submitted patch. The in-depth path shows the internal structure that makes those trajectories useful evidence: phase boundaries define what each run must produce, the agent chooses or synthesizes the tests during those phases, durable shared memory persists tests and handover state, deterministic tools make localization and edits inspectable, and cross-agent selection converts those independently chosen tests into the B_{ij} and R_{ij} signals used below.

6.1 Agent runtime contract

The agent runtime is specified declaratively: one configuration file defines the phase graph, prompt templates, action-interface family, prompt-guidance skill library, and operational limits. The runtime contract is therefore explicit before execution, instead of being recoverable only from scattered scripts.

Phases. The evaluated phase graph has eight nodes and fifteen directed edges (Algorithm 1). Each phase encodes a local workflow and two possible exits: successful completion or explicit give-up. This structure does not make the agent more capable by itself; rather, it prevents long trajectories from becoming unstructured conversations. The phase definitions and handover prompts make phase exits auditable: when a run leaves a phase, the next phase is selected from the configured edge and the handover memo records what evidence is carried forward. Concretely, PHASEHANDOVER extracts scripts, traces, tests, patch diffs, verification logs, and reports from the exiting phase; chooses either the configured next edge or the configured fallback edge based on the exit reason; writes a memo that summarizes carried evidence and open risks; and preserves scratch notes in shared state rather than in the submitted patch.

Proposition 1 (Workflow well-formedness). *In the observed Python headline trajectories, phase visits use the configured phase names and the run terminates either in final reporting or in an explicit runtime outcome.* *Argument.* The phase graph is declared in configuration, and the trajectory-derived phase summaries show complete coverage of the eight declared phases for the 495 persisted trajectories. We do not claim liveness—the agent is not guaranteed to solve every instance—only that the published run is auditable against the configured phase graph.

Skills (phase-gated, tool-gated). Skills are reusable prompt-guidance blocks, not learned model capabilities. Each skill gives the agent a small procedure or tool-use pattern, and is injected only when the current phase and available tools make it relevant. The current configuration contains 14 such phase- and tool-gated skills. This deliberately conservative injection policy shortens the

prompt and narrows the action space, which matters when a repair trajectory can span hundreds of turns.

Action formats. The action interface separates the semantic requirement—one executable action per turn—from model-specific syntax. The same runtime can therefore evaluate several model families without changing the agent’s state machine or tool semantics.

Tool sandbox. At runtime, the agent receives a fixed set of deterministic software-engineering operations. The current set covers dynamic line tracing, caller discovery, and guarded line editing. Keeping this set small made tool behavior inspectable and prevented the agent from depending on a large, brittle tool zoo.

Templates. Templates enforce the interaction contract: each turn contains an observable rationale field and exactly one action, and long observations are summarized with explicit elision rather than silently dropped. We treat that field as trajectory text, not as a faithful record of the model’s cognitive state; the goal is to make the observable interface stable.

Context compression and handover. State handover addresses a simple failure mode: facts stored only in the conversation can disappear when context is compressed. At phase changes, and when a phase grows too long, the agent writes a concise memo and continues with the relevant shared artifacts. The filesystem, not the raw conversation, becomes the durable memory.

Orchestra runtime. For the Python headline submission, each turn is processed by two roles. A planning role drafts the next step, and a formatting role rewrites only the executable action to satisfy the active interface. Both roles use the same locally hosted model. This division is a harness-level substitute for fine-tuning: the planner can express the repair strategy naturally, while the formatter normalizes only the command surface before execution, reducing formatting failures without changing model weights.

6.2 Software-engineering tool suite

The suite is deliberately small: **line tracing** records which statements and values appear under a reproduction test, turning localization into execution evidence; **caller tracing** shows the paths that reach suspect code and connects local symptoms to user-visible behavior; and **guarded editing** couples each edit to an expected target line, avoiding many free-form patch corruptions. The main policy is phase-gated injection: the agent sees only the tools and guidance relevant to its current phase, making tool availability part of the experimental contract.

6.3 Multi-cluster pipeline

The pipeline turns benchmark experimentation into a set of reusable stages: candidate generation, grading, data synthesis, optional post-training, test-time selection, and reporting. Each stage can run locally for debugging or as a scheduled cluster job. The practical contribution is reuse: operators can re-run only the stage under study while holding earlier artifacts fixed. This is what made repeated TTS@8 sweeps feasible in day-to-day development.

6.4 Model-agnostic serving

The serving layer decouples model identity, chat formatting, and execution sandbox. The same agent can run against local open-weight models, multi-server vLLM deployments, and remote APIs [21]. The backend inventory is reported with the artifact materials.

6.5 CI as the evaluation substrate

CI is the evaluation substrate, not just a test runner. It records the stage graph, the cluster target, the reusable artifacts, and the conditions under which expensive work should be skipped. The stage graph covers preparation, candidate generation, grading, synthesis, optional training, post-training benchmarking, selection, and reporting; reuse short-circuits let operators iterate on one stage without invalidating the rest. This follows standard CI practice of keeping frequent integration feedback cheap and repeatable [10, 11].

Reuse mode lets the team hold a trajectory bundle fixed while experimenting with selector weights, or hold an environment fixed while testing a new model. It is the operational difference between a benchmark script and an industrial evaluation pipeline.

7 Cross-Agent Test-Time Selection

A central design choice is to separate *candidate generation* from *final selection*. With $K = 8$ independent runs of the same locally hosted model, each run produces a candidate patch together with its own bug-revealing and regression-preservation tests. The candidate agent decides which tests exist before final selection; the cross-agent selector does not choose tests, but only cross-applies the archived test runners. Each run contributes two kinds of evidence: tests intended to expose the reported bug, and tests intended to guard nearby behavior against regressions. These tests are fixed before final selection, so the selector can compare candidate patches without accessing hidden benchmark tests. At final selection time, we form a $K \times K$ matrix in which entry (i, j) is the result of running run j 's archived tests against run i 's patch, converting candidate diversity into a selection signal using only evidence archived by the candidate runs.

Selector. For each candidate i we compute

$$\text{score}(i) = \frac{1}{K} \sum_{j=1}^K (w_B B_{ij} + w_R R_{ij}) \quad (1)$$

Here B_{ij} and R_{ij} are the pass rates obtained by the cross-application process above: B_{ij} is the pass rate of candidate i on run j 's bug-revealing (FAIL_TO_PASS) tests, and R_{ij} is the corresponding pass rate on its regression (PASS_TO_PASS) tests. The weights $w_B=0.3$ and $w_R=0.7$ are pragmatically selected fixed global constants in the submitted selector artifact; nevertheless, an artifact-only sweep over seven nearby weight pairs changes at most 35/495 selected candidates and 5/500 estimated outcomes. We then pick $\arg \max_i \text{score}(i)$, breaking ties by patch length. A tie means that at least two deduplicated patches are within 10^{-3} of the best score; this occurs on 304/495 covered instances. The selector summary is reproduced as Table 2.

Why cross-agent rather than oracle Best-of-K. We do not use the official benchmark tests at selection time; doing so would leak

Table 2: Cross-agent selector summary.

Quantity	Value	Quantity	Value
Expected (Verified)	500	w_B	0.3
Selected (covered)	495	w_R	0.7
Resolved (xcheck selector)	376	Tie epsilon	0.001
Score over expected (xcheck)	0.752	Tie-breaker	shortest_patch_raw
Resolved (order baseline)	362	Tie-break applied	304/495
Score over expected (order)	0.724	Mean unique patches	5.59/8
Tie instances	304	Matrix executions	21,635
Reordered by selector	302		

Table 3: Candidate-level TTS@8 decomposition.

Quantity	Interpretation	Value
Mean pass@1	Average over eight runs	67.7%
All-run solved	$r_i = 8$ scaffold-easy instances	234 / 500
Zero-run solved	$r_i = 0$ scaffold-hard instances	92 / 500
Marginal instances	$1 \leq r_i \leq 7$ selector-sensitive	174 / 500
Oracle pass@8	Any run resolves	408 / 500 = 81.6%
Cross-agent selector	Official-evaluator result	374 / 500 = 74.8%
Selector regret	Oracle minus submitted result	34 / 500 = 6.8 pp

evaluation information. The same constraint matters in production: a bad merge can cause significant downstream damage, so the system needs a selection signal before any hidden or production-only feedback is available. Instead, each run contributes the tests that its own agent chose or generated while solving the issue. The selector treats those archived tests as fixed evidence and cross-applies them to other runs' patches, turning trajectory diversity into selection evidence without any learned verifier or LLM judge. The matrix admits only archived suites with both `has_fail_to_pass` and `has_pass_to_pass`, so empty or invalid generated-test bundles are dropped before selection (12–21 drops per leg); candidate patches are deduplicated by content hash, and the 2/2,768 deduplicated patches with non-ok apply status are excluded from scoring. We do not infer flakiness or over-breadth from logs; after archive creation the selector reads only suite outcomes and apply status, never SWE-bench hidden-test results. Per-instance reordering data backs this up: on 302/495 instances the selector chose a candidate other than the order baseline, demonstrating that it actively re-ranks rather than acting as a no-op.

Diversity ceiling. A simple bound on selector quality comes from the union and intersection of the eight runs: at least one run resolves 408 instances, while all runs resolve 234. The cross-agent selector reaches 376 on the internal Docker re-grade (374 on the official evaluator), or about $376/408 = 92.2\%$ of the oracle union ceiling under the internal Docker re-grade. Pairwise completion overlap is high, so the useful diversity is mostly in *which* instances are solved, not in whether runs cover different parts of SWE-bench Verified. The candidate-level decomposition sharpens the bound: for 92 instances, none of the eight runs produces a resolving candidate, so these instances cannot be recovered by a better selector. Only 174 instances are “marginal”—resolved by some but not all runs ($1 \leq r_i \leq 7$)—and therefore actually expose a selection choice. Measured against the official Python headline, the cross-agent selector leaves 34 instances (6.8 pp) below the oracle union; linear interpolation on the closed-form oracle pass@k curve places the cross-agent selector near an oracle with $k \approx 2.2$, so much of the eight-way compute budget is currently spent on candidate generation that the selector does not fully monetize.

8 Evaluation

Evaluation setup. RQ1–RQ5 use SWE-bench Verified as the fixed Python task suite; RQ6 uses Multi-SWE-bench Java Verified as a cross-language validation track. The Python headline configuration uses a locally hosted model out of the box: the approach does not require training or fine-tuning before connecting the model to the agent runtime. For Python we run the same model eight times, archive each run’s candidate patch and tests, and then apply the cross-agent selector described in §7. We report the official SWE-bench cloud evaluation as the primary Python result and use internal Docker re-grading only as a consistency check. The Java RQ6 result uses the same 27 B agent and phase graph, but the Java xcheck@8 artifact and Java leaderboard rather than the Python Docker/cloud evaluator. This setup separates three questions for each track: how often the full system resolves issues, how it compares with published peers, and where remaining errors come from.

The Python analysis uses the published TTS@8 artifacts: 495 SWE-bench harness reports, 495 trajectories, the cross-agent selection bundle, and the per-instance outcome vectors of a curated competitor set. We adopt the canonical denominator $N=500$ for every SWE-bench Verified aggregate; the five trajectory/report pairs that are missing from the bundle are folded into the unresolved bucket as MISSING_ARTEFACT failures rather than dropped. The Java analysis uses the published Multi-SWE-bench Java Verified submission bundle with canonical denominator $N=128$, described separately in RQ6.

8.1 RQ1: Headline accuracy and uncertainty

Result. The Python configuration resolves 374/500 instances on the official cloud evaluator (74.80%, Wilson 95% CI [70.82%, 78.41%] [38]). The corresponding internal Docker re-grade resolves 376/500 (75.2%); the two-instance delta reflects evaluator differences and we report the official Python number as the Python headline. A repository-clustered bootstrap ($B=10,000$ replicates resampling at the repository level) returns a wider 95% CI of [67.0%, 79.8%], reflecting the fact that resolution is correlated within repositories; we therefore report both intervals to make the headline comparable to peers under their own reporting conventions and conservative to clustering.

Per-repository breakdown. Table 4 gives the repository breakdown. The largest repository, django/django, accounts for 46.2% of the Python benchmark and resolves at 76.6%, just above the global mean. The lowest-rate repository is pylint-dev/pylint at 30.0% ($n=10$). The auxiliary per-year CSV remains available in the artifact; across 2017–2023, year-level resolution stays in the 66–94% band, with the high end driven by the small 2017 slice. We report the structural repository pattern rather than mechanism because the workspace does not contain a controlled study of repository-internal API churn or year effects.

Threats. Python cross-system comparisons against published numbers from other research groups use heterogeneous evaluation stacks; we therefore restrict our Python headline number to the official evaluator and use paired per-instance outcome vectors (rather than aggregate rates) for all Python peer comparisons in RQ2. A run-to-run variance estimate beyond the eight-run sweep would

Table 4: SWE-bench Verified resolution by repository.

Repository	Resolved/Total	Rate	Repository	Resolved/Total	Rate
astropy/astropy	13/22	59.1%	pydata/xarray	18/22	81.8%
django/django	177/231	76.6%	pylint-dev/pylint	3/10	30.0%
matplotlib/matplotlib	23/34	67.7%	pytest-dev/pytest	16/19	84.2%
mwaskom/seaborn	1/2	50.0%	scikit-learn/scikit-learn	27/32	84.4%
pallets/flask	1/1	100.0%	sphinx-doc/sphinx	30/44	68.2%
psf/requests	8/8	100.0%	sympy/sympy	57/75	76.0%
Overall: 374/500 (74.8%)					

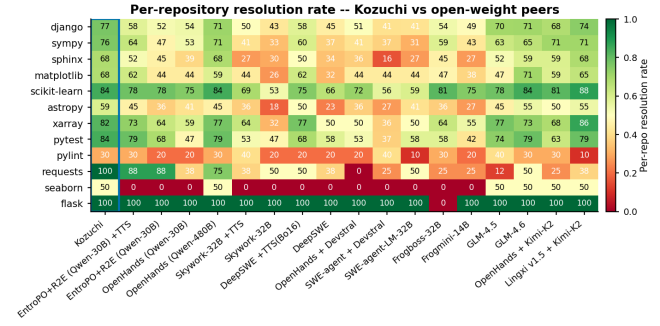


Figure 3: Per-repository resolution rate for Kozuchi agent (first column on the left) and the curated open-weight peers.

require additional eight-run sweeps under the same Orchestra configuration; we instead report the per-run pass@1 dispersion in RQ5.

8.2 RQ2: Peer-ranked position with multiplicity-corrected tests

Setup. We rank our system on a frozen snapshot of 135 publicly cataloged SWE-bench Verified as of April 28, 2026, and run paired McNemar exact tests [26] on the per-instance outcome vectors of a curated set of 17 open-weight and 7 closed-frontier peers. Multiplicity is controlled across the 24 paired tests with both Holm–Bonferroni (family-wise error rate) and Benjamini–Hochberg (false discovery rate) procedures [5, 16], each with target $\alpha = 0.05$ or $q = 0.05$.

Result. On the frozen Python leaderboard snapshot, the configuration ranks **12th overall** and is the highest-ranked open-weight system; the next open-weight submission, *Lingxi v1.5 + Kimi-K2*, resolves 356/500 (71.2%). Under Benjamini–Hochberg FDR control ($q \leq 0.05$), the configuration significantly outperforms 16 of 17 curated open-weight peers, including a 480B Qwen3-Coder configuration ($\Delta = +26$ instances, BH-FDR $q = 0.011$). The single peer the configuration does not significantly outperform under either FWER or FDR control is *Lingxi v1.5 + Kimi-K2* (raw $p = 0.054$, $\Delta = +18$). Among closed-frontier peers, the configuration is statistically indistinguishable at $\alpha = 0.05$ from *Sonar + Claude-Sonnet-4.5* ($\Delta = 0$), *OpenHands + Claude-Opus-4.5*, *live-SWE + Gemini-3-Pro*, and *OpenHands + GPT-5*; significantly outperforms *OpenHands + Claude-4-Sonnet* (FDR-significant); and is significantly behind the two top *Sonar / live-SWE + Claude-Opus-4.5* builds (FDR-significant; $\Delta = -22$ in both cases). The full pair-wise tables list raw McNemar p -values, Holm-corrected p -values, BH-corrected q -values, and conditional log-odds-ratio effect sizes with exact 95% Clopper–Pearson confidence intervals [8].

Per-repository peer structure. Figure 3 shows that the open-weight lead is not only an aggregate leaderboard artifact. On the largest benchmark slice, `django/django` ($n = 231$), Kozuchi agent resolves 76.6%, ahead of the next open-weight peer in the heatmap at 74.0%; on `sympy/sympy` ($n = 75$) it resolves 76.0%, compared with 70.7% for the strongest peers; and on `pytest-dev/pytest` ($n = 19$) it reaches 84.2%, above the nearest peers at 78.9%. The same pattern appears on several medium-size repositories: `sphinx-doc/sphinx` is tied for best at 68.2%, and `pydata/xarray` is near the top at 81.8%. The main exception is `pylint-dev/pylint`, where Kozuchi agent resolves only 30.0% (3/10) and several peers reach 40.0%. We therefore interpret the peer advantage as broad but not uniform: it is strongest on the large repositories that dominate the Verified denominator, while low-count repositories should be read as diagnostic failure clusters rather than stable rank evidence. §8.6 extends the peer position analysis to Java, where the same 27 B configuration ranks first among strict open-weight submissions.

Threats. Ranks use a frozen snapshot of cataloged submissions.

8.3 RQ3: Where does performance vary?

Setup. For the Python SWE-bench Verified run, we stratify resolution by patch size, by per-instance LLM-call count, and by phase-level events extracted from the 495 trajectory JSON files; we also report a multivariate logistic regression with cluster-robust standard errors (clustering at the repository level) that adjusts trajectory features for external consensus hardness proxies. Because the consensus terms use peer outcomes, we treat this model as an explanatory diagnostic rather than as a deployable predictor.

Patch size. Resolution rate decreases monotonically with the line-of-code (LOC) churn of the produced patch: from 81.7% at 1–4 LOC to 41.7% at >100 LOC. The point-biserial correlation between LOC churn and the binary resolution outcome is $r = -0.197$ ($p = 1.1 \times 10^{-5}$), the strongest single feature signal in the dataset. The mean LOC added per patch is 10.5 (median 3) for resolved instances versus 21.9 (median 8) for unresolved instances.

Trajectory effort. Trajectory-level effort metrics (per-instance LLM calls, runtime, prompt tokens, bash invocations) all carry *negative* correlations with resolution. We interpret this signal as a hardness-selection effect: harder instances cost more effort and are also more likely to fail. We do not interpret it as “effort hurts”; no effort metric is positively correlated with resolution at $\alpha = 0.05$, which we report as a structural ceiling on what further inference compute can buy at fixed scaffold and backbone.

Phase-level behaviour. All 495 trajectories visit every one of the eight phases; rework is concentrated in the `CODE_FIX` ↔ `VERIFY_PATCH` loop. `VERIFY_PATCH` issues a `GIVEUP` in 18.2% of trajectories (90/495, 95% CI [15.0%, 21.8%]); `CODE_FIX` itself rolls back in 1.0% of cases; every other phase rolls back zero times. Unresolved trajectories fire roughly 28% more `CODE_FIX` messages than resolved ones (mean 170 vs. 132) and report 0.67 vs. 0.52 mean `VERIFY_PATCH GIVEUPs` per instance.

Multivariate adjustment. A logistic regression of the binary resolution outcome on log-transformed LLM calls, patch churn, runtime, and two external consensus hardness proxies

Table 5: Failure taxonomy for unresolved instances.

Category	Count	Share
WRONG_FIX	115	91.3%
REGRESSION	6	4.8%
MISSING_ARTEFACT	5	4.0%
PATCH_DID_NOT_APPLY	0	0.0%
EMPTY_PATCH	0	0.0%
Total	126	100.0%

(`qwen_consensus` and `frontier_consensus`) shows that the consensus terms dominate the multivariate signal. After this adjustment, the trajectory-effort and patch-churn coefficients are not significant at $\alpha = 0.05$. This supports the hardness-selection interpretation above and prevents the univariate patch-churn correlation from being read as an independent causal predictor.

Threats. The consensus-adjusted model requires other systems’ outcomes, and its coefficients should not be read causally.

8.4 RQ4: Failure taxonomy and operational reliability

Setup. For the Python SWE-bench Verified run, we classify each unresolved instance into one of five mutually exclusive failure categories from the SWE-bench harness reports. We then report the operational profile of the 495 trajectories that produced a harness report: phase visit completeness, exit status, and patch-application reliability.

Failure taxonomy. Of 126 unresolved instances, 115 (91.3%) are `WRONG_FIX`: the patch applies cleanly under the harness but does not flip the hidden `FAIL_TO_PASS` tests. 6 (4.8%) are `REGRESSION`: the patch applies but breaks at least one `PASS_TO_PASS` test. 5 (4.0%) are `MISSING_ARTEFACT`: no trajectory or harness report was persisted for the instance under the published bundle. The `PATCH_DID_NOT_APPLY` and `EMPTY_PATCH` categories are both 0. The taxonomy is summarised in Table 5.

Operational reliability. All 495 trajectories report `exit_status = "Submitted"`; the agent does not crash mid-flight in this run. Of the 495 produced patches, 494 apply cleanly through the SWE-bench harness, giving a single edit-layer failure (0.20%). Combined with the failure taxonomy, this isolates the residual gap as a semantic-correctness problem rather than an edit-format problem: the agent essentially always emits a syntactically acceptable patch, but in 91.3% of unresolved cases the patch is semantically wrong.

Threats. `MISSING_ARTEFACT` cases are operational rather than modelling failures: under stricter persistence handling some of them might be recovered at no algorithmic cost. They are not removed from the Python headline denominator in this paper, but their existence inflates the unresolved bucket by up to 5/500 (1.0 pp).

8.5 RQ5: TTS@8 selection lift and cost-diversity envelope

Setup. For Python, we compare per-run pass@1 patches against the cross-agent TTS@8 selection on the same eight runs. Cost is measured in additional compute: 8× candidate generation plus the cross-agent test cross-application matrix. For Java, the shipped bundle exposes the xcheck-selected source-run counts rather than

Table 6: Artifact-based TTS@8 ablation.

Variant	Resolved	Rate	Note
$K=1$, no selector (mean)	338.6	67.7%	eight-run mean
$K=1$, no selector (best)	343	68.6%	best leg
$K=8$, order baseline	362	72.4%	no cross-agent ranking
$K=8$, self-tests only	346	69.2%	estimated
$K=8$, cross-agent selector	376	75.2%	selected
$K=8$, oracle	408	81.6%	upper bound

complete per-leg pass@1 reports, so we use those counts only as a source-run audit of the selected Java outcome.

Per-run pass@1. The pass@1/pass@k terminology follows standard code-generation evaluation practice [7]. The eight Python pass@1 runs are tightly clustered: they resolve 334–343 of 500 instances (66.8–68.6%), with mean $338.6/500 = 67.7\%$ and population standard deviation 3.12 resolved instances (0.62 pp). The submitted cross-agent selection raises this average-run baseline to $374/500 = 74.8\%$ on the official evaluator and $376/500 = 75.2\%$ on the internal Docker re-grade, a gain of 35.4–37.4 resolved instances (+7.1–7.5 pp) over mean pass@1. For Java, the xcheck artifact selects 15.9 source-run candidates on average and resolves 5.1 of them (population standard deviation 1.54), yielding $41/128 = 32.03\%$ overall and a 32.3% conditional resolved rate over selected source-run contributions.

Selection lift. On Python, the cross-agent selector resolves $376/500 = 75.2\%$ on the internal Docker re-grade (Table 2) and $374/500 = 74.8\%$ on the official evaluator. On 302 of 495 covered instances the selector chooses a different candidate than the size-ordered first candidate. The order baseline (first-run-wins) resolves $362/500 = 72.4\%$; the selector therefore contributes +14 absolute (+2.8 pp) over the order baseline and +33 to +42 absolute over individual per-run pass@1.

Selector ablation. Table 6 separates the measured Python TTS@8 gain into candidate generation and selection effects. Moving from a single candidate to eight candidates creates the 408/500 oracle ceiling, while the cross-agent selector captures 376 of those instances on the internal Docker re-grade. A diagnostic self-tests-only selector, computed from the same archived cross-agent selection tables by scoring each candidate only on its own generated tests and then checking the chosen run’s per-run report, resolves $346/500$. The gap between self-tests-only selection and the cross-agent selector’s $376/500$ result supports the specific value of cross-agent test application, while the oracle gap leaves 32 internal Docker re-grade wins for better selection at fixed candidates. This ablation isolates candidate count and selector signals; the other harness mechanisms below are supported by operational signatures rather than by controlled component-removal reruns.

Diversity ceiling. Across the eight Python runs the union of resolved instances is 408 and the intersection is 234. The selector’s 376 corresponds to about 92% of the internal Docker re-grade oracle ceiling (376/408). Pairwise Jaccard similarity over *completed* instances ranges from 0.943 to 0.965 (mean 0.957), so the diversity is in *which* instances are resolved rather than in coverage of the benchmark itself.

Compute envelope. On Python, TTS@8 buys the +14 absolute lift over the size-order baseline at $8 \times$ candidate generation plus a separately logged selector matrix: 21,635 patch–suite executions over 495 covered instances (mean 5.59 deduplicated patches $\times 7.80$

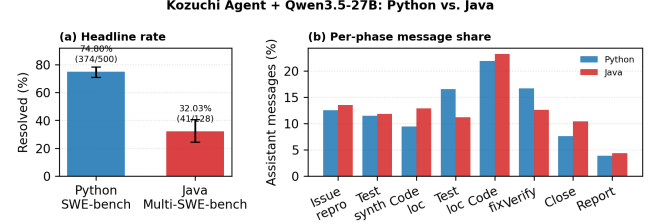


Figure 4: Cross-language transfer summary.

archived suites, versus a naive $8 \times 8 \times 495 = 31,680$ matrix), booked as 50 cross-check shards in `xcheck_manifest.json`. We measure candidate-generation cost from the published trajectories rather than from job logs. Across the 495 trajectories, per-instance medians are 490 LLM calls, 6.20M prompt tokens, 83.1k completion tokens, and 0.77 h of wall-clock; the p95 tail extends to 1,099 LLM calls, 16.6M prompt tokens, and 1.94 h. Aggregated over the full eight-run candidate generation, the configuration consumes approximately 3.8×10^9 prompt tokens and 5.8×10^7 completion tokens, so the workload is prompt-token dominated, consistent with the persistent handover and shared-state mechanism described in §6. The compute–resolution Pareto curve over the 374 resolved instances shows diminishing returns in the long tail: 80% of resolved instances finish within 653 LLM calls per instance and 95% within 1,069, but moving from the 90% point to the 99% point requires increasing the budget from 820 to 1,547 calls. We report this Pareto as a descriptive cost curve over *this* run rather than as a prescription; an early-stopping policy that exploits it would need its own controlled study.

Threats. Cross-agent testing requires that each candidate run produces high-quality agent-generated tests; weak tests directly weaken the selector’s discrimination. A controlled comparison of direct versus debate-style selection on the same trajectory bundle would require a paired selector run with otherwise identical inputs. The compute envelope is reported in LLM calls, prompt/completion tokens, and wall-clock seconds rather than in GPU-hours, because the trajectory bundle does not record GPU-utilisation traces; converting to GPU-hours requires cluster-side accounting we did not collect. The ablation in Table 6 isolates candidate count and selector signals, but it does not isolate the causal contribution of the phase graph, persistent handover, action formatter, tracing tools, or guarded editing; those components require separate controlled reruns. The auditable selector boundary is the archived suite outcome and patch apply-status table, so hidden SWE-bench outcomes are unavailable until after patch selection.

8.6 RQ6: Cross-language transfer to Multi-SWE-bench Java

We re-ran the same Qwen3.5-27B agent, eight-phase scaffold, and $K=8$ selection pattern on Multi-SWE-bench Java Verified. Only the corpus and harness change: Java uses Maven/Gradle and strict `xcheck@8`, while the Python headline uses `pytest` and TTS@8 (§7). The phase graph, action contract, persistent state, and selection family remain the same. Figure 4 shows headline resolution with Wilson intervals [38] and per-phase assistant-message share.

Table 7: Python–Java cross-track evidence for the same 27 B Kozuchi agent configuration.

Signal	Python	Java
Benchmark	SWE-bench Verified (n=500)	Multi-SWE-bench Java Verified (n=128)
Resolved (Wilson 95% CI)	374/500 = 74.80% [70.82, 78.41]	41/128 = 32.03% [24.57, 40.54]
Leaderboard position	12/135 overall	4/42 overall
Open-weight rank	1st curated open-weight	1st strict open-weight
BH-FDR peer wins	16/17 open-weight peers	34/41 Java peers
Same-family open margin	+26 vs. Qwen3-Coder-480B baseline	+38 vs. Qwen2.5-72B CODE_FIX +1.3 pp; VERIFY_PATCH -4.1 pp
Rework-loop phase delta		
Median LLM calls	490	529
Nonempty applying patch output	494/495 = 99.80%	124/128 = 96.9%
Globally novel solves	1	1

The Java run resolves 41/128 instances (32.03%, Wilson 95% CI [24.57%, 40.54%]), ranks 4 of 42 overall, and is first among strict open-weight submissions. Under BH-FDR control ($q \leq 0.05$), it significantly outperforms 34 of 41 Java peers, including all 72 B Qwen2.5 open-weight builds ($\Delta = +27$ to $+38$). The only FDR-significant adverse comparison is *CodeArts-Agent* + *CodeArts-MiniMax-M2.5* ($\Delta = -15$, $q = 0.011$), with a much larger closed-weight backbone.

The scaffold behaviour transfers more strongly than the absolute score: per-phase message share is within ± 5 pp on every phase, rework still centres on CODE_FIX $\leftrightarrow$ VERIFY_PATCH, and six of seven effort medians sit within $\pm 20\%$ of Python parity. Thus the 43 pp headline gap is not a compute gap. The Java failure profile is instead language/harness-specific: no_fix_test_results (25%), report_valid_not_leaderboard_resolved (9%), and anomalous_test_pattern (6%) do not appear on Python; edit-layer reliability is lower (96.9% vs. 99.8%); and elastic/logstash accounts for 38/128 Java instances, with 0/38 solved by Kozuchi agent and 32/38 unresolved by every catalogued peer. We therefore claim language-agnostic harness transfer, not equal language fluency.

The Java leaderboard mixes open-weight, semi-open, undeclared, and closed-frontier peers; our open-weight claim uses the strict “decision-LLM weights publicly released” subset. As in RQ5, the Java run does not separately ablate scaffold, backbone, and selector contributions.

9 Production Planning and Lessons Learned

The deployment lessons are operational rather than architectural. First, phase boundaries beat free-form planning at long horizons: every Python TTS@8 trajectory visits all eight phases, rework concentrates in CODE_FIX $\leftrightarrow$ VERIFY_PATCH, and the same phase shape transfers to Java within ± 5 percentage points per phase (§8.6). Second, model-family variation is cheapest to handle at the action-format boundary: switching tool-call syntax is a configuration change rather than an agent fork. Third, CI reuse is a core evaluation lever, because operators can hold trajectories, selection bundles, or environments fixed while changing one stage. The same principle applies to filesystem-backed state: reproduction notes, generated tests, traces, and handover memos survive context compression and make runs reviewable after the fact. Finally, cross-agent testing turns candidate diversity into accuracy without a learned verifier, while one cluster abstraction keeps multi-site runs reviewable.

The artifact-derived operational indicators are consistent with those lessons: the workflow compresses five operator touch-points

to one CI push (about 70 engineer-minutes saved per cycle), the runtime exposes four action formats across 17 model configs, six of nine CI stages are reusable, and five cluster environments sit behind ENV_NAME. On Python, 494/495 produced patches apply cleanly and the cloud-vs-Docker score drift is only 374 vs. 376 out of 500. On Java, the same runtime contract reaches 41/128 (32.03%) and ranks first among strict open-weight submissions, making cross-language transfer a measured operational outcome rather than an anecdotal consistency check.

10 Conclusion

Production plan. Kozuchi Agent targets conservative production use on large repositories, with evidence from SWE-bench Verified (Python) and Multi-SWE-bench Java Verified. Using a locally hosted open-weight model, no fine-tuning, eight candidate runs, and cross-agent testing, it resolves 374/500 SWE-bench Verified instances (74.80%, Wilson 95% CI [70.82%, 78.41%]); repository-clustered bootstrap [67.0%, 79.8%]) and 41/128 Multi-SWE-bench Java Verified instances (32.03%) under the same agent, backbone, and scaffold. It ranks 12th among 135 catalogued Python submissions, is the highest-ranked open-weight Python system, ranks first among strict open-weight Java submissions, and recovers 80% of resolved Python instances within 653 LLM calls.

What we learned. The results point to harness engineering—phase decomposition, single-command grammar, persistent shared filesystem, deterministic SE tools, cross-agent test selection, cluster indirection, and CI reuse—while ablations isolate only candidate count and selector signals. The phase decomposition, action grammar, persistent state, and cross-agent selector transfer from Python to Java unchanged, supporting the language-agnostic claim. The selector adds +14 resolved Python instances; 494/495 Python patches apply cleanly; and remaining Python errors are mainly semantic or selection failures: 91.3% of unresolved instances have clean-applying patches that miss hidden tests, and 34 instances are solved by some run but not selected.

Next. Future work should ablate tools, selector design, and post-training; use conversation audits for content-aware selection; connect failure-pattern detection to harness updates; and use the compute-resolution Pareto curve for controlled early stopping. Industrial-impact claims remain limited to workflow replacement, cross-evaluator consistency, and the Python–Java benchmark evidence; Java cross-track results are first-class evidence rather than a single consistency point, while controlled pilots and broader multilingual benchmarks remain future work.

11 Artifact Availability

The source index for the references below is maintained in paper/final/source-index.md; it records the URL and paper-specific use of each external source.

The public artifact is the SWE-bench Verified submission folder at <https://github.com/kimusaku/experiments>. It contains the 495 trajectories (trajs/), harness reports (logs/), predictions (all_preds.jsonl), cross-agent selection bundle (xcheck/), and released analysis scripts under src/. From these files alone, src/build.sh regenerates the empirical RQ1–RQ5 values; the Java submission directory under the same artifact family regenerates

the RQ6 values. All RQ figures, trajectory/conversation audits, and the selector re-run over the eight published candidate streams are included in that artifact family.

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